



Department of Computer Science and Engineering
National Institute of Technology, Rourkela
Mid-Semester Examination (Spring 2025 - 26)
MTech 1st Semester
Sub: Foundations of Computer Security (CS - 6301)
Time: 2 Hrs; FM: 30

Answer **ALL** questions.

1. (a) List and explain the attacks that threaten the integrity. [3] [CO1, BL1]
(b) Distinguish between passive and active security attacks. List a few of them under each category. [2] [CO1, BL3]
2. (a) Using the Euclidean algorithm, find the gcd of 14 and 320. [3] [CO2, BL2]
(b) Using the properties of the mod operator, prove that the remainder of any integer when divided by 1000 is the integer made of the three rightmost digits. [2] [CO2, BL3]
3. (a) Find the multiplicative inverse of 7 in Z_{180} using the extended Euclidean algorithm. [2] [CO1, BL2]
(b) Find the solution of a linear Diophantine equation $40x + 30y = 88$. [3] [CO3, BL2]
4. (a) Find all solutions of the linear equation $232x + 42 \equiv 248 \pmod{50}$ [2] [CO3, BL2]
(b) Find the solutions to the linear equation
$$\begin{aligned} 2x + 3y &\equiv 5 \pmod{8} \\ x + 6y &\equiv 3 \pmod{8} \end{aligned}$$
[3] [CO2, BL3]
5. (a) Encrypt the message “**this is an exercise**” using the Additive cipher with key $K = 20$ and decrypt the message to get the original plaintext. [5] [CO2, BL3]
(b) Use a Hill cipher to encipher the message “We live in an insecure world”. Use the following key: [5] [CO2, BL3]

$$K = \begin{bmatrix} 03 & 02 \\ 05 & 07 \end{bmatrix}$$